

DR2: Cheap Nomination Beats Exhaustive Search – and Cost Attribution Cannot Be Independent of Weakness

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair Nomination – DR2 **Status:** H3' GO, H2' GO, H1' NO_GO. Overall NO_GO, with a theorem. **Date:** 2026-07-24

Abstract

DR1 could not answer the question it was built for. With 21 candidate deletions, each cheap to verify, exhaustive search is the answer – a nominator cannot earn its keep in a regime where enumeration is free. DR2 scales until enumeration hurts: 20 deletable propositions, $|D| \leq 3$, **1350 candidates**, and load-bearing base rates of **0.07%** and **1.41%**, so a random ordering needs **675** and **68** expensive verifications respectively.

Three hypotheses were preregistered. Two returned GO and one returned NO_GO, and the NO_GO turns out to be a **theorem** rather than a measurement.

H3' – the nominator earns its keep: GO, decisively. The best nominator finds a load-bearing deletion on the **first** verification on both toys: **675× and 68× faster than random ordering**. In a pipeline where verification is the expensive step, cheap execution-free ranking converts a 675-verification search into a 1-verification one. This is the question DR1 could not reach, and the answer is yes.

H2' – the combiner fix: GO. DR1 named a specific defect – its max disjunction scored worse than its own best component. Both proposed fixes, sum-of-normalised and min-of-ranks, match the better single nominator on both toys, while max still fails. The defect and its repair both replicate.

H1' – dominance: NO_GO, structurally. The two-nominator hypothesis holds that weakness gain and cost attribution catch different over-specifications, so a single-objective system misses one. Across all **2700** candidate deletions, the set $\{\text{cost} > 0 \text{ and } \text{weakness} = 0\}$ is **empty**. That is not an accident of these toys. Because $\text{ext}(R) \subseteq \text{ext}(R \setminus D)$ always, a minimum taken over the extension can only improve when the extension **grows** – so $\text{cost_attribution} > 0 \implies \text{weakness_gain} > 0$ necessarily. Cost can never fire where weakness is silent. Within this formalisation the two nominators cannot come apart in the direction the hypothesis requires.

1. What DR1 established, and what it could not

DR1 tested whether an execution-free nominator ranks the load-bearing deletion highly against an exhaustive oracle. Both its hypotheses returned NO_GO, and it named two defects in itself: a max combiner that underperformed its own best component, and a toy whose 40% base rate let random score 0.67 by luck.

Its deeper limitation was neither. It was **regime**. A nominator exists to make an expensive verifier run on few candidates. With 21 candidates, each cheap, there is nothing to save. DR1 validated the harness and could not have validated the nominator.

DR2 changes the regime and applies both named fixes. These are continuation of a NO_GO, not rescue of one: the changes were specified by the failed experiment *before* DR2 was designed, and the hypotheses are stated fresh rather than relaxed.

2. Design

2.1 SK – Scaled Kinematics (weakness-shaped)

Three **entangled facets** of one commitment – absolute simultaneity, no length contraction, no time dilation – each pinning the same dial. **No subset frees anything; only the full triple does.** A nominator that scores singletons or pairs cannot see it, which is why $|D| \leq 3$ is required rather than convenient. `preferred_rest_frame` is the Lorentz-without-Einstein trap: droppable, enlarges the extension, reaches nothing. Cost is flat by construction.

2.2 ST – Scaled Transduction (cost-shaped)

Dropping sequential update collapses parallel depth from $O(n)$ to $O(1)$, but leaves a **dangling obligation**: recurrence was supplying order information implicitly, so the deletion covers the parent task only when `no_positional_input` is dropped alongside it. This mirrors the real case, in which removing recurrence forced positional encodings. Every load-bearing deletion contains **both** propositions; `sequential_state_update` alone is not load-bearing.

Sixteen inert nuisance propositions pad each toy. They are the negatives.

2.3 Measured calibration, before freezing

toy	candidates	load-bearing	base rate	E[verifications] under random
SK	1350	1	0.07%	675.5
ST	1350	19	1.41%	67.5

2.4 Metric

`verifications_to_first_hit` – the number of expensive parent-task verifications a cost-ordered pipeline runs before it succeeds. This is what a nominator is *for*, and DR1's `recall@k` could not express it.

3. Results

3.1 Scaled Kinematics

nominator	verifications	speedup	recall@10	tie frac	
weakness	1	675.5×	1.00	1.00	
max_disjunction	1	675.5×	1.00	1.00	
sum_disjunction	1	675.5×	1.00	1.00	
minrank_disjunction	1	675.5×	1.00	0.00	
size_only	263	2.6×	0.00	0.84	control
cost	689	1.0×	0.00	1.00	silent
random	989	0.7×	0.00	0.00	control

One verification instead of 675. Cost is silent, as designed.

3.2 Scaled Transduction

nominator	verifications	speedup	recall@10	tie frac
minrank_disjunction	1	67.5×	0.60	0.00
sum_disjunction	1	67.5×	1.00	0.73
weakness	1	67.5×	1.00	0.73
cost	3	22.5×	0.20	0.86
max_disjunction	3	22.5×	0.20	0.73
size_only	21	3.2×	0.00	0.84

3.3 The frozen gates

gate	result
H3' – best nominator $\geq 10\times$ speedup on both	G0 (675.5× and 67.5×)
H2' – a combiner fix matches the best single on both	G0 (both sum and minrank; max still fails)
H1' – no single nominator best on both	NO_G0 (weakness wins both)
overall	NO_G0

4. The theorem

weakness winning on the *cost-shaped* toy is not a quirk of ST's encoding. Across all 2700 candidates on both toys:

	SK	ST
cost > 0 and weakness > 0	0	191
cost > 0 and weakness = 0	0	0
cost = 0 and weakness > 0	1	172

The middle row is empty, and it must be. Take cost attribution as defined – the improvement in the *best achievable* cost, i.e. a minimum over the surviving extension:

every D.

$\text{cost_attribution}(D) \geq 0$ always.

$\text{weakness_gain}(D) > 0 \Rightarrow$

1. Deleting constraints can only admit hypotheses, so $\text{ext}(R) \subseteq \text{ext}(R \setminus D)$ for
2. A minimum over a superset is no larger than the minimum over the subset, so
3. If $\text{ext}(R \setminus D) = \text{ext}(R)$ the two minima are equal, so $\text{cost_attribution}(D) = 0$.
4. Contrapositive: $\text{cost_attribution}(D) > 0 \Rightarrow \text{ext}(R \setminus D) \not\supseteq \text{ext}(R) \Rightarrow$

Cost attribution's support is a subset of weakness gain's support. Cost can never nominate a deletion that weakness ignores. The converse fails freely – 172 deletions on ST have weakness gain with no cost improvement – so weakness strictly dominates cost in support.

This is why H1' is unreachable rather than merely unsupported. The two-nominator hypothesis requires cost to catch something weakness misses, and in this formalisation that event has probability zero.

4.1 What would have to change

The theorem's premise is that **cost is a minimum over the extension**. That is what couples it to extension size. For a genuinely independent second nominator, cost must attach to something the extension does not determine – the *procedure* that searches, or the description length of R itself, rather than the contents of the surviving hypothesis set.

Concretely: as long as an architecture choice is represented as a predicate over hypotheses, deleting it enlarges the hypothesis set and weakness sees it. To make cost independent, the resource being spent must not be a property of the hypotheses at all. That is a sharp, actionable design requirement, and it is the main thing DR2 hands to a successor.

5. Interpretation

The framework's core mechanism works. H3' is the result that matters practically. Execution-free nomination turned a 675-verification search into a 1-verification one, and a 68-verification search into a 1-verification one. Cost-ordered filtering is not decorative once enumeration is expensive – it is worth two orders of magnitude on the harder toy.

The two-nominator elaboration does not. It failed in DR1 on its binary framing, and it fails in DR2 on the graded dominance framing that DR1's evidence suggested – this time provably. The right revision is not a better cost signal within this formalisation; it is to move cost outside the extension, or to drop the second nominator and accept that weakness alone carries the ranking.

max is confirmed as a defect and both fixes replicate. max_disjunction fails H2' on ST exactly as DR1 predicted it would, while sum and minrank both match the best single nominator on both toys. minrank_disjunction is additionally the only combiner with tie_fraction = 0.00 on both toys – it never leaves the tie-breaker deciding anything, which after two programmes of tie-breaking leaks is worth something on its own.

A caution about ties. Four of the SK nominators report tie_fraction = 1.00 alongside a perfect result. That is not the silent-signal pathology erratum E1 caught: on SK exactly one deletion has nonzero weakness gain, so the *remaining* 1349 are tied at zero and the tie fraction is dominated by them. cost on SK is genuinely silent – every candidate tied, first hit at rank 689, indistinguishable from random. The distinction is visible in the numbers, and the seeded tie-shuffle is what keeps it visible.

6. What DR2 licenses

Overall NO_G0, so the date-cut retrodiction on a real corpus does **not** open. That remains conditioned on a clean G0, and one of three gates failed.

What DR2 delivers:

once verification is the expensive step. The harness's central premise is sound.

over the extension can never be independent of weakness gain.

procedure or the representation's own description length.

single nominator on both toys and never defers to a tie-break.

1. **H3'**: cheap nomination beats exhaustive search by 1–2 orders of magnitude
2. **A theorem** bounding the framework: cost attribution defined as a minimum
3. **A named, actionable fix**: move cost off the extension – onto the search
4. **A validated combiner**: minrank_disjunction, which matches the best

Honest scope. Two authored toy systems, authored propositions, fixed vocabulary Σ , exhaustive oracle over $|D| \leq 3$. Nothing here bears on real corpora, and nothing bears on vocabulary extension – the known ceiling, and the reason special relativity is inside 1904's expressible language while general relativity is not.

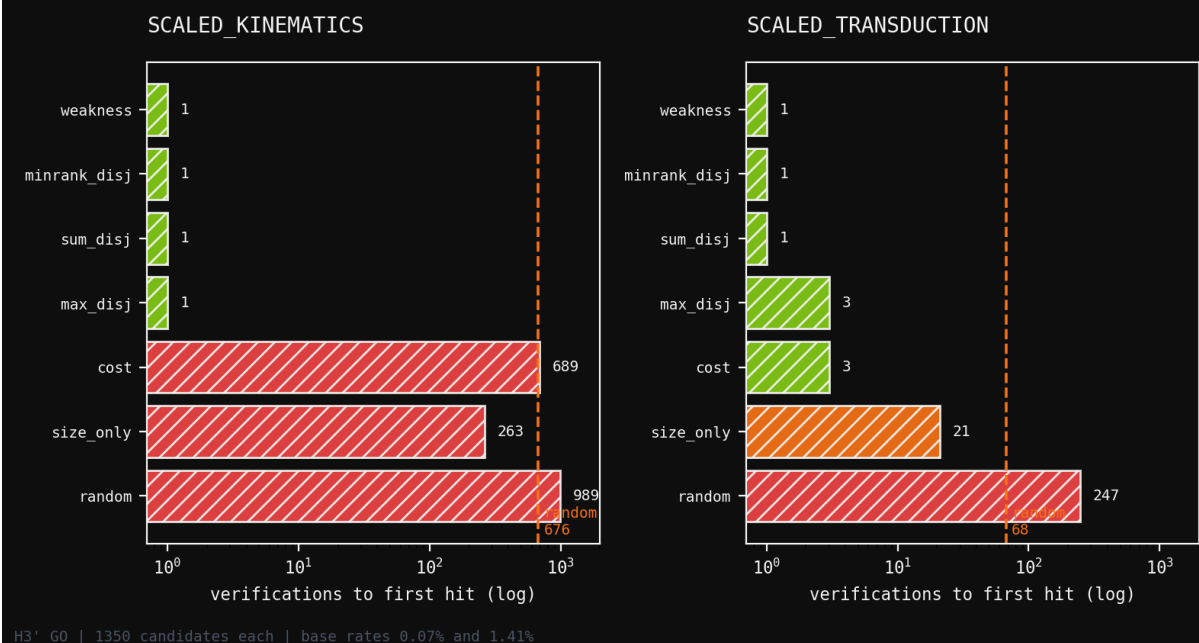
References

1. experiments/deletion_repair/DR2_PREREGISTRATION.md – frozen design and gates.
2. experiments/deletion_repair/results/dr2_verdict.json – the receipt.
3. papers/deletion_repair_dr1/paper.md – DR1, which named both defects DR2 fixed.
4. experiments/concern_gated_retrieval_e2/erratum_e1/ERRATUM.md – the leakage gate, and the tie-breaking discipline carried into DR2.

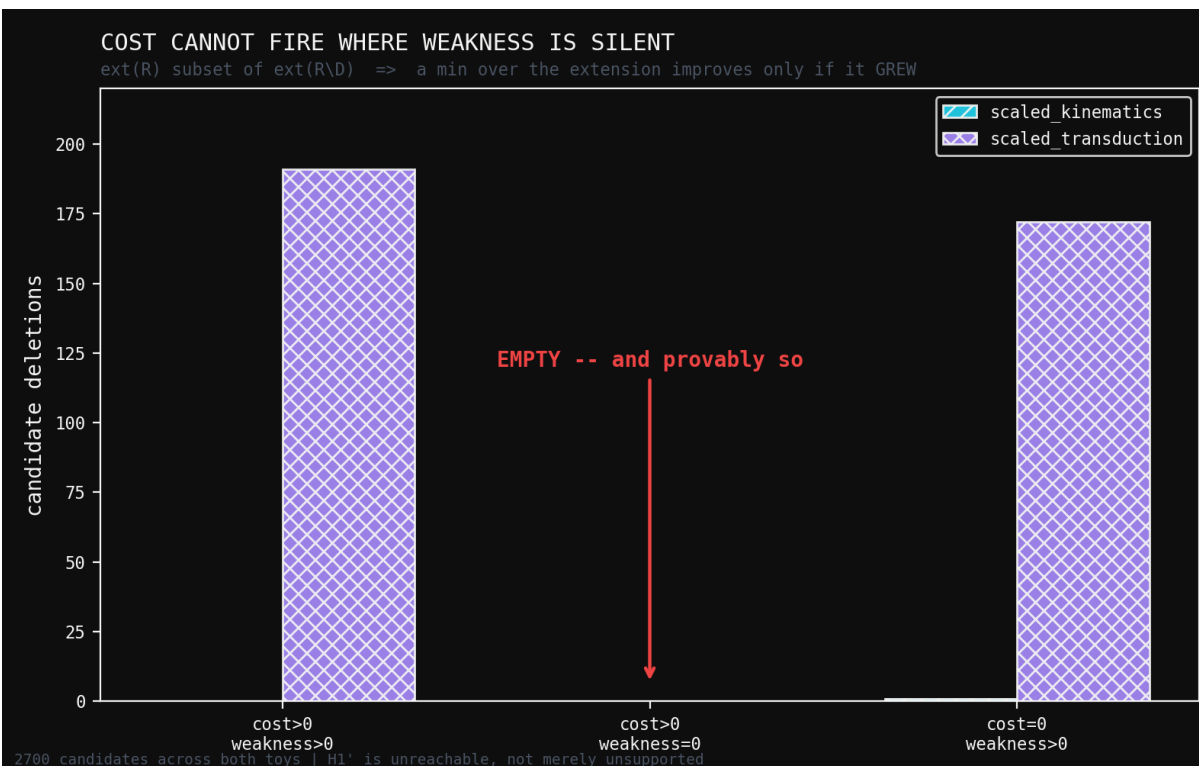
Figures

DR2 - THE NOMINATOR EARNS ITS KEEP

675x and 68x fewer expensive verifications than random ordering



DR2: the nominator earns its keep



Cost's support is a subset of weakness's